

Kartic

Ph.D. Candidate, Life Sciences · Gachon University, Seongnam-si

+82-10-3884-0847 hi.kartic@gmail.com ORCID 0009-0005-5939-4192 github.com/kartic03

Personal Information

Nationality Indian

Address Seongnam-si, Gyeonggi-do, Republic of Korea

Education

2023. 03 ~ 2026. 12 **Gachon University**, Seongnam-si, Korea · Ph.D. in Life Sciences
GPA 4.25 / 4.5 · Advisor: Prof. Tae-Sik Park, Dean of Research
Thesis · AI-Driven Peptide Engineering: A De Novo Design Cascade for Drug Delivery Across the Blood–Brain Barrier
Defence December 2026; degree conferral February 2027

2019 ~ 2021 **Shoolini University**, Solan, India · M.Sc. in Biotechnology
CGPA 8.64 / 10

2016 ~ 2019 **Shoolini University**, Solan, India · B.Sc. (Hons.) in Biotechnology
CGPA 8.03 / 10
Shoolini University ranks 452 globally, QS World University Rankings 2027, and first among private universities in India

Research Experience

2023. 03 ~ present **Lipidomics and Metabolic Disease Laboratory**, Department of Life Sciences, Gachon University · Graduate Researcher
Principal Investigator: Prof. Tae-Sik Park
Doctoral research on de novo peptide design for blood–brain barrier transport, spanning computational design and wet-lab validation. Duties detailed below.

Technical Skills

PROGRAMMING	Python · R · C/C++ · SQL
FRAMEWORKS AND LIBRARIES	PyTorch · TensorFlow · scikit-learn · Biopython · scikit-bio
PROTEIN ML	ESM-2 · ESMC · sparse autoencoders · transformers · retrieval-augmented and multi-task architectures
PROTEIN DESIGN STRUCTURE	RFdiffusion · ProteinMPNN · AlphaFold2 / AlphaFold3 · Boltz-1 / Boltz-2 · ESMFold · ColabFold Schrödinger suite · molecular docking · structure-based virtual screening · molecular dynamics simulation
STATISTICS	Nested cross-validation · leakage auditing · calibration and uncertainty · applicability domain · preregistration · equivalence testing
SEQUENCING	FastQC · fastp · BWA · Bowtie2 · STAR · Salmon · featureCounts · DESeq2 · GATK · bcftools · IGV
CELL CULTURE	Human iPSC and ReN neural progenitors · astrocyte and neuronal differentiation · SH-SY5Y · HEK293 · HepG2 · AML12
IMAGING	Fluorescence microscopy · Zeiss LSM 980 confocal (certified) · immunofluorescence · live-cell imaging · colocalisation · quantitative image analysis
MOLECULAR	Restriction and Gibson cloning · plasmid and vector design · PCR, RT-PCR and quantitative PCR · DNA/RNA isolation · electrophoresis and gel purification

PROTEIN	Protein expression in E. coli · bacterial culture and selection · affinity chromatography · SDS-PAGE · western blotting · DLS and zeta potential for size, surface charge and oligomeric state of proteins, protein complexes and protein–nucleic acid complexes
SPOKEN LANGUAGES	English (fluent) · Hindi (native)

Core Competencies

■ Machine learning for protein and peptide design

- De novo peptide design cascade: RFDiffusion backbone generation, ProteinMPNN sequence design, validation with AlphaFold2/3 and Boltz-1/2
- Protein language models (ESM-2, ESMC), sparse autoencoders for feature-level attribution, retrieval-augmented and multi-task architectures in PyTorch
- Structure-based virtual screening, molecular docking and molecular dynamics (Schrödinger)
 - Able to take a design task from representation choice to a shortlist worth synthesising, and to state which candidates the model is entitled to make a claim about.

■ Model validity: leakage auditing, calibration and applicability domain

- Nested cross-validation, calibration and uncertainty quantification, applicability-domain characterisation
- Preregistration and equivalence testing, so that a negative result is a defensible claim rather than a failure to detect
 - Able to build the evaluation layer that tells a group when a model has left the region it was trained on, before that failure costs an experimental cycle.

■ Cell biology and imaging

- Human iPSC and ReN neural progenitors; astrocyte and neuronal differentiation; immortalised lines (SH-SY5Y, HEK293, HepG2, AML12)
- Zeiss LSM 980 confocal (certified), immunofluorescence, live-cell imaging, colocalisation and quantitative image analysis
 - Able to test a computational prediction in the cell type it is supposed to act on, and to generate the imaging data rather than wait for someone else to.

■ Protein expression and biophysical characterisation

- Protein expression in E. coli, bacterial culture, affinity chromatography, SDS-PAGE, western blotting
- DLS and zeta potential for hydrodynamic size, surface charge and oligomeric state of proteins, protein complexes and protein–nucleic acid complexes
- Restriction and Gibson cloning, vector design, PCR, nucleic acid isolation
 - Able to express and characterise a designed sequence in-house, closing the loop between a model's ranking and a measurable property.

Research Duties

■ Doctoral project: de novo peptide design for blood–brain barrier transport

- Built a hybrid sequence–structure machine-learning framework predicting BBB-penetrant peptides, published in *Brain Network Disorders* (first author, 2026)
- Designed the full generation-to-validation cascade and carried it through to cell-based testing
 - Sole first author; designed, implemented and validated the pipeline, and wrote the manuscript.

■ Benchmark and audit methodology

- ToxBench: a leakage-audited multi-task toxicology benchmark reporting calibration, uncertainty and applicability domain alongside accuracy (*BMC Bioinformatics*, 2026)
- Preregistered, equivalence-tested audit of acquisition shortcuts in EEG foundation models across five public corpora (*Scientific Reports*, minor revision)
 - Established the evaluation protocol before analysis in both studies, which is what made the negative findings reportable.

■ Clinical-data modelling in Parkinson's disease

- Cognitive-decline modelling in the PPMI cohort using cerebrospinal fluid markers (*Diagnostic and Prognostic Research*, submitted)
- Explainable prediction of recorded deep brain stimulation status, with cross-dataset biomarker analyses (*BMC Medical Informatics and Decision Making*, revision submitted)
 - Handled data governance, statistical design and reporting; every computational manuscript ships with a public repository.

■ Laboratory contributions

- Co-authored wet-lab studies on hepatic lipogenesis, recombinant cell-permeable protein transduction, and natural-product pharmacology
- Wrote and maintained the group's analysis pipelines in Python and R
 - Bridged the computational and experimental sides of the group, supporting colleagues' projects with analysis as well as running my own.

Peer-reviewed Publications

1. ToxBench: a leakage-audited multi-task benchmark for predictive toxicology with calibration, uncertainty, and applicability domain analysis

Kartic, Yeeun Seo, Sanggyun Yi, Tae-Sik Park

BMC Bioinformatics, 2026 **FIRST AUTHOR** doi:10.1186/s12859-026-06621-x

2. Blood-brain barrier-transport artificial intelligence: a hybrid sequence-structure machine-learning framework for predicting blood-brain barrier-penetrant peptides

Kartic, Amitesh Sharma, Sanggyun Yi, Tae-Sik Park

Brain Network Disorders, 2026 **FIRST AUTHOR** doi:10.1016/j.bnd.2026.04.001

3. Inhibition of hepatic lipogenesis and adipogenesis by cordyanhydride A isolated from *Cordyceps militaris* cultivated on germinated soybeans

Kartic, Amitesh Sharma, Nattan Stalin, Sanggyun Yi, Amna Parveen, Sunyeou Kim, Tae-Sik Park

Bioscience, Biotechnology, and Biochemistry, 2026 **FIRST AUTHOR** doi:10.1093/bbb/zbag058

4. Recombinant cell-permeable puromycin *N*-acetyltransferase confers puromycin resistance by direct protein transduction

Jiwon Choi, Kyung-Hee Cho, Jiwon Im, Yeeun Seo, Amitesh Sharma, **Kartic**, Shivani Devi, Nattan Stalin, Seo Jin Park, Tae-Sik Park

Journal of Microbiology and Biotechnology, 2025 doi:10.4014/jmb.2502.02049

5. The ethanolic extract of *Chrysanthemum morifolium* Ramat flower activates parasympathomimetic and anti-inflammatory effects for prevention of presbyopia

Amitesh Sharma, Shivani Devi, **Kartic**, Nattan Stalin, Jiwon Lim, Jiwon Choi, Jisun Kim, Mi-Young Jeong, Sung-Joon Lee, Tae-Sik Park

Natural Product Sciences, 2024 doi:10.20307/nps.2024.30.1.20

Publications Under Review

1. **RATAN-PBind: retrieval-augmented, target-aware nomination of de novo protein binders within a characterized applicability domain**
Kartic, Jiwon Choi, Tae-Sik Park
Journal of Cheminformatics **FIRST AUTHOR** UNDER REVIEW
2. **A preregistered equivalence-tested protocol for auditing acquisition shortcuts in EEG foundation models**
Kartic, Shivani Devi, Yeeun Seo, Sanggyun Yi, Tae-Sik Park
Scientific Reports **FIRST AUTHOR** MINOR REVISION
3. **Where cerebrospinal fluid markers do and do not predict cognitive decline in early Parkinson's disease**
Kartic, Shivani Devi, Sanggyun Yi, Yeeun Seo, Tae-Sik Park
Diagnostic and Prognostic Research **FIRST AUTHOR** SUBMITTED
4. **Explainable prediction of recorded deep brain stimulation status in Parkinson's disease: a preliminary retrospective model with exploratory cross-dataset biomarker analyses**
Kartic, Jiwon Lim, Tae-Sik Park
BMC Medical Informatics and Decision Making **FIRST AUTHOR** REVISION SUBMITTED

Conferences

2024. 05	KSBMB International Conference, Busan, Korea
2024. 05	3rd Korea–Japan Lipid Joint Symposium, KSBMB, Busan, Korea
2023. 10	Lipidomics Perspectives on Metabolic Diseases, Seoul National University, Korea

Certifications and Training

2026. 09	Cyclic Peptide Modeling and Informatics · Schrödinger
2026. 03	AI Fluency: Framework and Foundations · Anthropic
2026. 03	Claude Code in Action · Anthropic
2025. 08	Confocal Microscopy, Zeiss LSM 980 · Yonsei University
2022. 06	Environment and Sustainability · workshop, IIIT and JUIT
2022. 03	Advances in Characterization Techniques and Applications: A Material Science Perspective · faculty development programme, Ramaiah Institute of Technology with Shriram Institute of Industrial Research
2022. 01	Shapes Effect on Hydrogen Response of Palladium Nanoparticles · webinar, IEC University
2020. 02	Introduction to Professional Scientific Communication · NPTEL, IIT Kanpur

Awards

2013. 02	National Cyber Olympiad, 3rd position · Science Olympiad Foundation
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